# ASSIGNMENT 9B

**Name:** Rohit Kailas Deore
**Roll No:** 11
**Batch:** T1
**Class:** TE-IT

[1]:
```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

## 1. Load the Mall Customers Dataset

[2]:
```
# Load the Mall Customers dataset from a direct CSV link
# Try more than one source in case a repository link changes.
urls = [
    "https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/
  ↪master/Mall_Customers.csv",
    "https://raw.githubusercontent.com/tirthajyoti/Machine-Learning-with-Python/
  ↪master/Datasets/Mall_Customers.csv",
]

df = None
for url in urls:
    try:
        df = pd.read_csv(url)
        print("Dataset loaded successfully.")
        break
    except Exception:
        print("Could not load this source. Trying the backup source...")

if df is None:
    raise RuntimeError(
        "Could not download the dataset. Check your internet connection, "
        "or download Mall_Customers.csv and load it using pd.
  ↪read_csv('Mall_Customers.csv')."
    )

print("First five rows:")
display(df.head())
print("Dataset shape:", df.shape)
```

```
Dataset loaded successfully.
First five rows:
```

| | CustomerID | Genre | Age | Annual Income (k$) | Spending Score (1-100) |
|---|---|---|---|---|---|
| 0 | 1 | Male | 19 | 15 | 39 |
| 1 | 2 | Male | 21 | 15 | 81 |
| 2 | 3 | Female | 20 | 16 | 6 |
| 3 | 4 | Female | 23 | 16 | 77 |
| 4 | 5 | Female | 31 | 17 | 40 |

```
Dataset shape: (200, 5)
```

## 2. Select Features

[3]:
```
# Use Annual Income and Spending Score for clustering
X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

print("Features used for clustering:")
display(X.head())
```

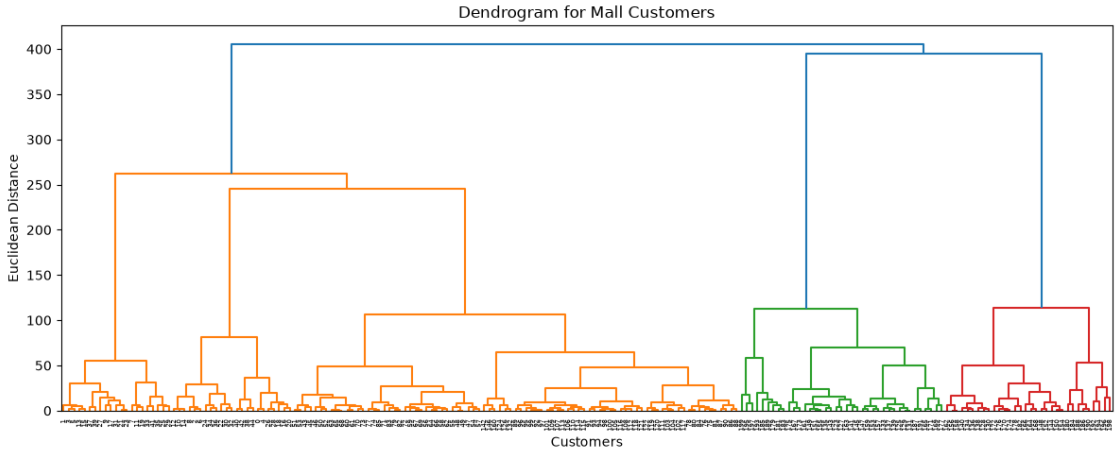
```
Features used for clustering:
```

| | Annual Income (k$) | Spending Score (1-100) |
|---|---|---|
| 0 | 15 | 39 |
| 1 | 15 | 81 |
| 2 | 16 | 6 |
| 3 | 16 | 77 |
| 4 | 17 | 40 |

## 3. Draw a Dendrogram

[4]:
```
# Ward linkage is commonly used for hierarchical clustering
linked = linkage(X, method="ward")

plt.figure(figsize=(12, 5))
dendrogram(linked)
plt.title("Dendrogram for Mall Customers")
plt.xlabel("Customers")
plt.ylabel("Euclidean Distance")
plt.tight_layout()
plt.show()
```




## 4. Apply Agglomerative Clustering

[5]:
```
# Use 5 clusters for comparison with the K-Means practical
model = AgglomerativeClustering(n_clusters=5, linkage="ward")
df["Cluster"] = model.fit_predict(X)

print("Hierarchical clustering completed with 5 clusters.")
display(df[["Annual Income (k$)", "Spending Score (1-100)", "Cluster"]].head())
```

```
Hierarchical clustering completed with 5 clusters.
```

| | Annual Income (k$) | Spending Score (1-100) | Cluster |
|---|---|---|---|
| 0 | 15 | 39 | 4 |
| 1 | 15 | 81 | 3 |
| 2 | 16 | 6 | 4 |
| 3 | 16 | 77 | 3 |
| 4 | 17 | 40 | 4 |

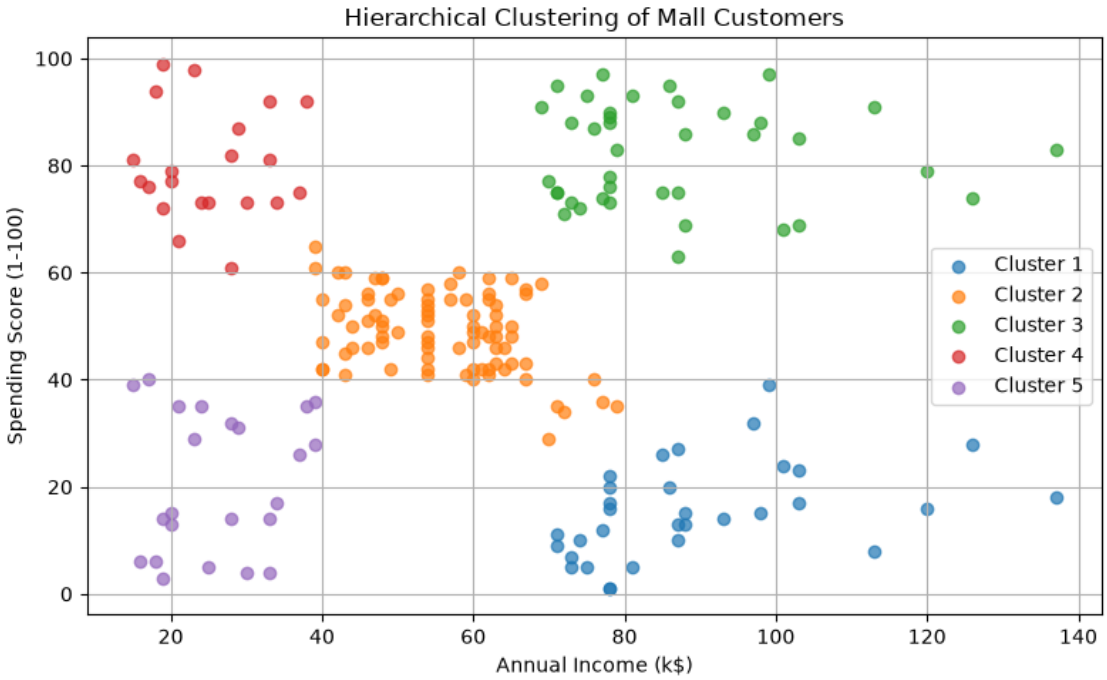
## 5. Visualize the Clusters

[6]:
```
plt.figure(figsize=(8, 5))

for cluster in sorted(df["Cluster"].unique()):
    cluster_data = df[df["Cluster"] == cluster]
    plt.scatter(
        cluster_data["Annual Income (k$)"],
        cluster_data["Spending Score (1-100)"],
        label=f"Cluster {cluster + 1}",
        alpha=0.7
    )

plt.title("Hierarchical Clustering of Mall Customers")
```

```
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```




## 6. Cluster Analysis

[7]:
```
summary = df.groupby("Cluster").agg(
    Number_of_Customers=("Cluster", "size"),
    Average_Annual_Income=("Annual Income (k$)", "mean"),
    Average_Spending_Score=("Spending Score (1-100)", "mean")
).round(2)

display(summary)
```

| Cluster | Number_of_Customers | Average_Annual_Income | Average_Spending_Score |
|---|---|---|---|
| 0 | 32 | 89.41 | 15.59 |
| 1 | 85 | 55.81 | 49.13 |
| 2 | 39 | 86.54 | 82.13 |
| 3 | 21 | 25.10 | 80.05 |
| 4 | 23 | 26.30 | 20.91 |

## Conclusion

Hierarchical clustering was applied to the Mall Customers dataset using annual income and spending score. A dendrogram was plotted to show the hierarchical relationships, and Agglomerative Clustering grouped the customers into five clusters. The scatter plot and summary table help interpret the resulting groups.